

1	(a) (i)	gravitational force provides/is the centripetal force	B1	
		same gravitational force (by Newton III)	B1	[2]
	(ii)	$\omega = 2\pi / T$		
		$= 2\pi / (4.0 \times 365 \times 24 \times 3600)$	C1	
		$= 5.0 \text{ (4.98)} \times 10^{-8} \text{ rad s}^{-1}$	A1	[2]
	(b) (i)	(centripetal force =) $M_A d \omega^2 = M_B (2.8 \times 10^8 - d) \omega^2$		
		or		
		$M_A d_A = M_B d_B$	C1	
		$M_A / M_B = 3.0 = (2.8 \times 10^8 - d) / d$	C1	
		$d = 7.0 \times 10^7 \text{ km}$	A1	[3]
	(ii)	$GM_A M_B / (2.8 \times 10^{11})^2 = M_A d \omega^2$	B1	
		$M_B = (2.8 \times 10^{11})^2 \times d \omega^2 / G$		
		$= (2.8 \times 10^{11})^2 \times (7.0 \times 10^{10}) \times (4.98 \times 10^{-8})^2 / (6.67 \times 10^{-11})$	C1	
		$= 2.0 \times 10^{29} \text{ kg}$	A1	[3]